Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/25/2015
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Spaceflight Effects on Neurocognitive Performance: Extent, Longevity, and Neural Bases		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Gainesville	**State:**	FL
Zip Code:	32611-8205	**Congressional District:**	3
Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	08/26/2011	**End Date:**	07/13/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Loerch, Linda	**Contact Phone:**	
Contact Email:	linda.loerch-1@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 7/13/2017 due to transfer to PI's new institution (Ed., 1/11/2018) NOTE: Extended to 3/31/2019 per NSSC information and PI (Ed., 8/20/2012)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Bloomberg, Jacob (NASA Johnson Space Center) Mulavara, Ajitkumar (Universities Space Research Association) Wood, Scott (NASA Johnson Space Center)		
Grant/Contract No.:	NNX11AR02G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NRA NNJ10ZSA003N requested proposals to assess changes in elemental neurocognitive functions such as perception, motor control, memory, attention, language, executive function, and emotional processing following long duration spaceflight using both behavioral assessments and monitoring technologies such as functional magnetic resonance imaging (fMRI). In response to this call, we propose to perform structural and functional MR brain imaging to identify the relationship between changes in crewmember neurocognitive function and neural structural alterations following a six month International Space Station (ISS) mission. Our central hypothesis is that measures of brain structure, function, and network integrity will change from pre to post flight in crewmembers (Aim 1). Moreover, we predict that these changes will correlate with indices of cognitive, sensory, and motor function in a neuroanatomically selective fashion (Aim 2). Our interdisciplinary approach utilizes cutting edge neuroimaging techniques and a broad ranging battery of sensory, motor, and cognitive assessments that will be conducted pre flight, during flight, and post flight to investigate neuroplastic and maladaptive brain changes in crewmembers following long duration spaceflight. Success in this endeavor would 1) result in identification of the underlying neural mechanisms and operational risks of spaceflight-induced changes in behavior, and 2) identify whether a return to normative behavioral function following re-adaptation to Earth's gravitational environment is associated with a restitution of brain structure and function or instead is supported by substitution with compensatory brain processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this project will have relevance not only to understanding the effects of spaceflight on the human brain and behavior, but also for delineating the capacity of the brain to remodel in response to adaptive stimuli. As such, the results should prove informative for understanding the neural mechanisms associated with adaptive behavioral change and the rehabilitation of these changes during recovery periods.
Task Progress:	Year 1. The project received partial funding during the first selection year "Definition Phase." During the first year "Definition Phase," the team was very productive. We attained Institutional Review Board approval from both the University of Michigan and NASA, which means that the project risks and benefits have been vetted through independent review boards. We also created and had our Experiment Document (ED) approved by the ISSMP CCB. The ED is a technical report that outlines all of the methods we will utilize for this project. It is referred to throughout the course of the study by numerous groups as the "gold standard" for the experiments. The project received "select for flight" on May 30, 2012. This means that we have undergone a feasibility review to confirm that the experiments can be implemented with crewmembers on the International Space Station. In this project, we will measure various aspects of motor, sensory, and cognitive function in crewmembers (and a group of control subjects) at multiple points prior to, during, and following spaceflight. Measures of brain structure and function will also be obtained pre and post flight. During the first project year, we have been purchasing, setting up, and debugging various hardware components with which to obtain these measurements. We have also been writing, refining, and testing our software for running these experiments. In the final months of the first funding year, we ran the initial pilot subjects through our behavioral and brain imaging protocols. This will enable us to ensure that we are obtaining good, clean data from which we can calculate our measures of interest prior to study initiation. Year 2 During year 2 we continued to test and refine our measures. Our complementary, NSBRI-funded bed rest study is up and running with four subjects enrolled to date. We will utilize the same measures in the current study. This means that we have our data collection and analysis pipelines in place and ready to support flight data acquisition. Moreover, our trouble shooting has resulted in our selection of a head tapper device to stimulate the functional vestibular cortex in the MRI scanner, as opposed to our initial plan of using the auditory click method. The clicks need to delivered at a high decibel which can be somewhat unpleasant for subjects. We also obtained all of the necessary approvals for our inflight testing software, and it has been delivered for implementation with the next ISS software build. The PI has obtained both WebMirage and Launchpad accounts for data transfer. In addition, one crewmember has given their informed consent to participate in this study for increment 39-40. This project also includes a ground-based normative control study. We will initiate testing for this arm of the project within approximately the next month. Year 3 This past year was very successful with the completion of data collection for the bed rest (n=18) and ambulatory control (n=16) groups. We are in the process of analyzing both data sets and preparing manuscripts. The results will greatly enhance our interpretation of flight study data. Given this ability to leverage data collection from our ongoing NSBRI-funded bed rest study, we have extensively refined our data collection and analysis approaches. We have obtained all of our necessary ethical, hardware, and software approvals. Our first flight participant recently returned from the ISS and we have completed the first post flight data collection session. We have pre tested a few additional flight subjects and are anticipating smooth data collection and analysis processes given our experience with the bed rest and normative control subjects.
Bibliography Type:	Description: (Last Updated: 03/12/2026)
Articles in Peer-reviewed Journals	Koppelmans V, Hirsiger S, Mérillat S, Jäncke L, Seidler RD. "Cerebellar gray and white matter volume and their relation with age and manual motor performance in healthy older adults." Hum Brain Mapp. 2015 Jun;36(6):2352-63. Epub 2015 Feb 19. http://dx.doi.org/10.1002/hbm.22775 ; PubMed PMID: 25704867 , Jun-2015